# Quizrr Chapter-Wise Test for JEE Main - 2025

**Q.1**

The mean radius of earth is $\mathrm{R}$, its angular speed on its own axis is $\omega$ and the acceleration due to gravity at earth's surface is $\mathrm{g}$. What will be the radius of the orbit of a geostationary satellite?

**(A)** $\left(R^2 g / \omega^2\right)^{1/3}$

**(B)** $\left(R g / \omega^2\right)^{1/3}$

**(C)** $\left(R^2 \omega^2 / g\right)^{1/3}$

**(D)** $\left(R^2 g / \omega\right)^{1/3}$

**Q.2**

A satellite of mass $\mathrm{m}$ revolves around the earth of radius $\mathrm{R}$ at a height ' $x$ ' from its surface. If $g$ is the acceleration due to gravity on the surface of the earth, the orbital speed of the satellite is

**(A)** $\frac{g R^2}{R+x}$

**(B)** $\frac{\mathrm{gR}}{\mathrm{R}-\mathrm{x}}$

**(C)** $\mathrm{gx}$

**(D)** $\left(\frac{g R^2}{R+x}\right)^{1/2}$

**Q.3**

A satellite of mass $\mathrm{M}$ is moving in a circle of radius $\mathrm{R}$ under a centripetal force given by $\left(-\mathrm{k} / \mathrm{R}^2\right)$, where $\mathrm{k}$ is a constant. Then

**(A)** The kinetic energy of the particle is $\frac{\mathrm{k}}{12}\mathrm{R}$

**(B)** The total energy of the particle is $\left(-\frac{\mathrm{k}}{2\mathrm{R}}\right)$

**(C)** The kinetic energy of the particle is $\left(-\frac{\mathrm{k}}{\mathrm{R}}\right)$

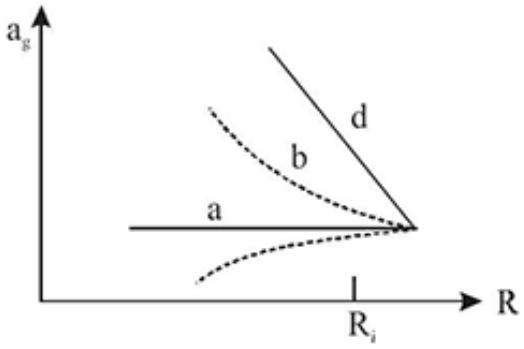
**(D)** The potential energy of the particle is $\left(\frac{\mathrm{k}}{2\mathrm{R}}\right)$

**Q.4**

A (nonrotating) star collapses onto itself from an initial radius $\mathrm{R_i}$ with its mass remaining unchanged. Which curve in figure best gives the gravitational acceleration $\mathrm{a_g}$ on the surface of the star as a function of

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the radius of the star during the collapse




**(A)** a

**(B)** b

**(C)** c

**(D)** d

**Q.5**

A satellite is launched into a circular orbit of radius R around the earth. A second satellite is launched into an orbit of radius $1.01R$. The time period of the second satellite is larger than that of the first one by approximately.

**(A)** 0.5

**(B)** 1.0

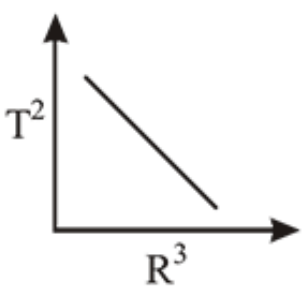
**(C)** 1.5

**(D)** 2.0

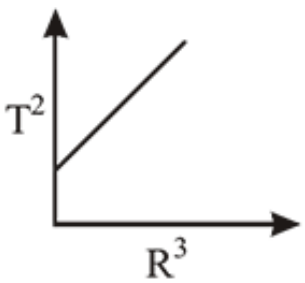
**Q.6**

Which of the following graphs represents the motion of a planet moving about the sun ?

**(A)**

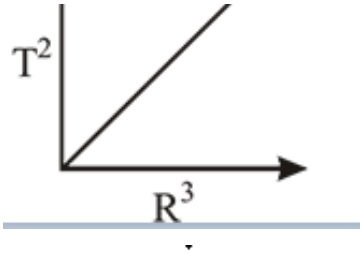



**(B)**

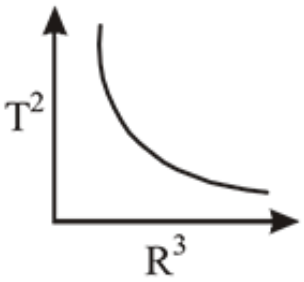



**(C)**

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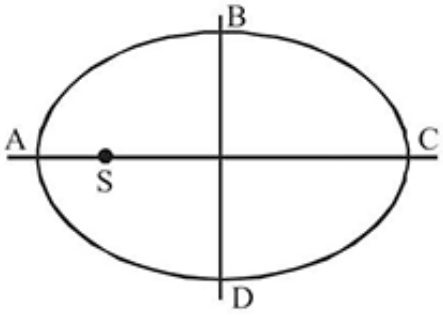



**(D)**




**Q.7**

A Planet is revolving around the sun.




Which of the following is correct option?

**(A)** The time taken in travelling $DAB$ is less than that for $BCD$

**(B)** The time taken in travelling DAB is greater than that for $BCD$

**(C)** The time taken in travelling CDA is less than that for $ABC$

**(D)** The time taken in travelling CDA is greater than that for $ABC$

**Q.8**

The time taken by the earth to travel over half of its orbit, remote from the sun, separated by the minor axis is about 2 days more than half the year, then the eccentricity of the orbit is

**(A)** $1/30$

**(B)** $1/60$

**(C)** $1/15$

**(D)** $1/70$

**Q.9**

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If the earth is treated as a sphere of radius $\mathrm{R}$ and mass $\mathrm{M}$; its angular momentum about the axis of its rotation with period $\mathrm{T}$, is

**(A)** $\frac{\pi \mathrm{MR}^3}{\mathrm{T}}$

**(B)** $\frac{\mathrm{MR}^2 \pi}{\mathrm{T}}$

**(C)** $\frac{2\pi \mathrm{MR}^2}{5\ \mathrm{T}}$

**(D)** $\frac{4\pi \mathrm{MR}^2}{5\ \mathrm{T}}$

**Q.10**

The period of moon's rotation around the earth is nearly 29 days. If moon's mass were 2 fold its present value and all other things remain unchanged, then what will be the period (in days) of moon's rotation ?

**(A)** 27

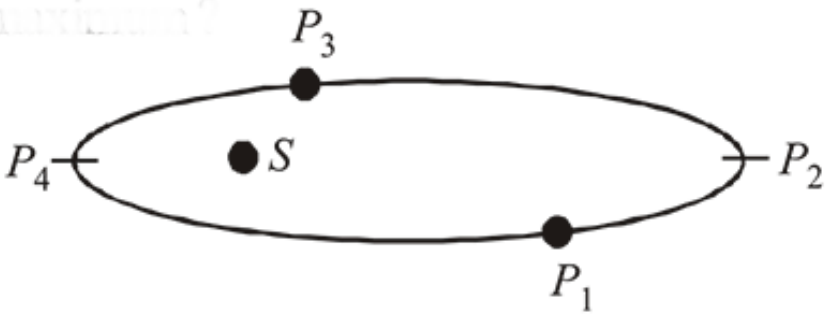
**(B)** 28

**(C)** 29

**(D)** 30

**Q.11**

The figure shows a planet in elliptical orbit around the sun $S$. Where is the kinetic energy of the planet maximum? $P_3$




**(A)** $P_1$

**(B)** $P_2$

**(C)** $P_3$

**(D)** $P_4$

**Q.12**

A planet revolves about the sun in elliptical orbit.
The areal velocity $\left(\frac{dA}{dt}\right)$ of the planet is $4.0 \times 10^{16}\ \mathrm{m}^2/\mathrm{s}$. The least distance between planet and the sun is $2 \times 10^{12}\ \mathrm{m}$. Then the maximum
speed of the planet in $\mathrm{km/s}$ is

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**(A)** 10

**(B)** 20

**(C)** 30

**(D)** 40

**Q.13**

In planetary motion the areal velocity of position vector of a planet depends on angular velocity $(\omega)$ and the distance of the planet from sun $(\mathrm{r})$. If so, the correct relation for areal velocity is

**(A)** $\frac{dA}{dt} \propto \omega r$

**(B)** $\frac{dA}{dt} \propto \omega^2 r$

**(C)** $\frac{dA}{dt} \propto \omega r^2$

**(D)** $\frac{dA}{dt} \propto \sqrt{\omega r}$

**Q.14**

The period of moon's rotation around the earth is nearly 29 days. If moon's mass were 2 fold its present value and all other things remain unchanged, the period of moon's rotation would be nearly

**(A)** $29\sqrt{2}$ days

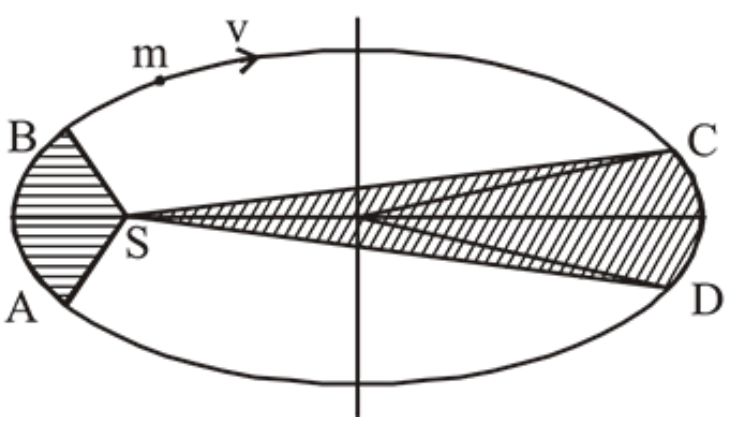
**(B)** $29/\sqrt{2}$ days

**(C)** $29 \times 2$ days

**(D)** 29 days

**Q.15**

The figure shows elliptical orbit of a planet $\mathrm{m}$ about the sun $\mathrm{S}$. The shaded area SCD is twice the shaded area $\mathrm{SAB}$. If $\mathrm{t}_1$ is the time for the planet to move from $\mathrm{C}$ to $\mathrm{D}$ and $\mathrm{t}_2$ is the time to move from $A$ to $B$ then




**(A)** $\mathrm{t}_1 = 4\mathrm{t}_2$

**(B)** $\mathrm{t}_1 = 2\mathrm{t}_2$

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**(C)** $t_1 = t_2$

**(D)** $t_1 > t_2$

**Q.16**

If the distance of earth is halved from the sun, then the no. of days in a year will be

**(A)** 365

**(B)** $182.5$

**(C)** 730

**(D)** 129

**Q.17**

If the earth is at one-fourth of its present distance from the sun, the duration of the year will be

**(A)** half the present year

**(B)** one-eighth the present year

**(C)** one-sixth the present year

**(D)** one-tenth the present year

**Q.18**

The distance of Neptune and Saturn from the sun is nearly $10^{13}$ and $10^{12}$ meter respectively. Assuming that they move in circular orbits, their periodic times will be in the ratio

**(A)** 10

**(B)** 100

**(C)** $10\sqrt{10}$

**(D)** 1000

**Q.19**

A satellite $A$ of mass $m$ is at a distance of $r$ from the surface of the earth. Another satellite $B$ of mass $2\ m$ is at a distance of $2r$ from the earth's centre. Their time periods are in the ratio of

**(A)** $1 : 2$

**(B)** $1 : 16$

**(C)** $1 : 32$

**(D)** $1 : 2\sqrt{2}$

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**Q.20**

Two satellites $A$ and $B$, having ratio of masses $3 \ : \ 1$ are in circular orbits of radius $r$ and $4r$. Calculate the ratio of total mechanical energies of $A$ to $B$.

**(A)** $\frac{12}{3}$

**(B)** $\frac{21}{1}$

**(C)** $\frac{11}{12}$

**(D)** $\frac{12}{1}$

## Answers & Solutions

**Q.1 Answer:**

$\left(R^2 g/\omega^2\right)^{1/3}$

**Solution:**

$T = \frac{2\pi r}{v_0} = \frac{2\pi r}{(gR^2/r)^{1/2}} = \frac{2\pi r^{3/2}}{\sqrt{gR^2}} = \frac{2\pi}{\omega}$

**Hence,** $r^{3/2} = \frac{\sqrt{gR^2}}{\omega}$ **or** $r^3 = \frac{gR^2}{\omega^2}$

**or,** $r = \left(gR^2/\omega^2\right)^{1/3}$

**Q.2 Answer:**

$\left(\frac{gR^2}{R+x}\right)^{1/2}$

**Solution:**

$\frac{\mathrm{mv}^2}{(\mathrm{R+x})} = \frac{\mathrm{GmM}}{(\mathrm{R+x})^2}$ **also** $\mathrm{g} = \frac{\mathrm{GM}}{\mathrm{R}^2}$

$\therefore \frac{\mathrm{mv}^2}{(\mathrm{R+x})} = \mathrm{m}\left(\frac{\mathrm{GM}}{\mathrm{R}^2}\right)\frac{\mathrm{R}^2}{(\mathrm{R+x})^2}$

$\therefore \frac{\mathrm{mv}^2}{(\mathrm{R+x})} = \mathrm{mg}\frac{\mathrm{R}^2}{(\mathrm{R+x})^2}$

$\therefore \mathrm{v}^2 = \frac{\mathrm{gR}^2}{\mathrm{R+x}} \Rightarrow \mathrm{v} = \left(\frac{\mathrm{gR}^2}{\mathrm{R+x}}\right)^{1/2}$

**Q.3 Answer:**

The total energy of the particle is $\left(-\frac{\mathrm{k}}{2\mathrm{R}}\right)$

**Solution:**

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$\frac{mv^2}{R} = \frac{k}{R^2}$ or $mv^2 = \frac{k}{R}$

**Kinetic energy** $= \frac{1}{2}mv^2 = \frac{k}{2R}$

**In case of satellites** $P.E = -2\ K.E$

**and** $T.E = P.E + K.E$

**Total energy** $= \frac{k}{2R} - \frac{k}{R} = -\frac{k}{2R}$

**Q.4 Answer:**

b

**Solution:**

$g \propto \frac{1}{R^2}$

**R decreasing $g$ increase hence, curve $b$ represents correct variation.**

**Q.5 Answer:**

1.5

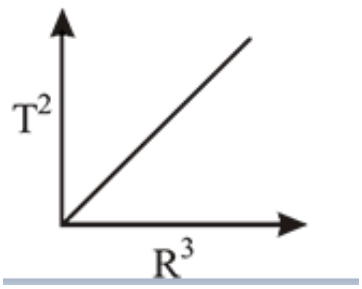
**Solution:**

$T = 2\pi\sqrt{\frac{(R+h)^3}{GM}}$

$T_1 = 2\pi\sqrt{\frac{R^3}{GM}}, \quad T_2 = 2\pi\sqrt{\frac{(1.01R)^3}{GM}}$

$\frac{T_2 - T_1}{T_1} \times 100 = 1.5\%$

**Q.6 Answer:**




**Solution:**

**No solution available**

**Q.7 Answer:**

The time taken in travelling $DAB$ is less than that for $BCD$

**Solution:**

**When closer to the sun, velocity of planet will be greater. So time taken in covering a given area will be less.**

**Q.8 Answer:**

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1/60

**Solution:**

$\frac{t}{\text{year}} = \frac{\text{area of sectional}}{\text{qrea of ellipse}}$

$\frac{\frac{\pi ab}{2}+eab}{t} = \frac{\pi ab}{2year}$

$\Rightarrow t = \frac{\frac{\pi ab}{2}+eab}{\pi ab}$ **yeqr**

$t = \frac{1}{2} + \frac{e}{\pi}$ **year** $= \frac{1}{2}$ **year** $+ \frac{e}{\pi}$ **year**

$\frac{e}{\pi}$ **year** $= 2day$

$\frac{e}{\pi} = \frac{2}{365}$

$e = \frac{2\times 3.14}{365}$

$e = \frac{1}{60}$

**Q.9 Answer:**

$\frac{4\pi \mathrm{MR}^2}{5\ \mathrm{T}}$

**Solution:**

**Angular momentum,** $\mathrm{L} = \mathrm{I}()$**; moment of inertia of sphere along the axis passing through centre of mass,**

$\mathrm{I} = \frac{2}{5}\mathrm{MR}^2$ **and** $\omega = \frac{2\pi}{\mathrm{T}}$

**Putting these values,** $\mathrm{L} = \frac{4\pi \mathrm{MR}^2}{5\ \mathrm{T}}$

**Q.10 Answer:**

29

**Solution:**

**Time period does not depend upon the mass of satellite**

**Q.11 Answer:**

$P_4$

**Solution:**

**Angular momentum of the planet about** $S$ **is conserved. So,** $mvr =$ **constant.**

$v$ **is maximum when** $r$ **is minimum. So,** $v$ **is maximum at point** $P_4$**. Hence K.E. is maximum at** $P_4$**.**

**Q.12 Answer:**

40

**Solution:**

$\frac{L}{2m} = \frac{dA}{dt}$ $(L =$ **angular momentum** $)$

$\frac{mv_{\max}r_{\min}}{2m} = \frac{dA}{dt}; v_{\max} = \frac{2dA/dt}{r_{\min}} = 40$

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**Q.13 Answer:**

$\frac{dA}{dt} \propto \omega r^2$

**Solution:**

$\frac{\mathrm{dA}}{\mathrm{dt}} = \frac{\mathrm{L}}{2\,\mathrm{m}} \Rightarrow \frac{\mathrm{dA}}{\mathrm{dt}} \propto \mathrm{vr} \propto \omega \mathrm{r}^2$

**Q.14 Answer:**

29 days

**Solution:**

**The Period is given by** $\mathrm{T} = \frac{2\pi \mathrm{r}^{3/2}}{\sqrt{\mathrm{GM}}} \ldots\ldots (1)$

**where** $\mathrm{G}$ **= gravitational constant**

$\mathrm{M}$ **= mass of the earth**

**The period does not depend on the mass of the moon.**

$\mathrm{T} \propto \mathrm{r}^{3/2}$

**So, the period of the moon's rotation the earth remains constant when we change the mass of the moon.**

**The period is 29days**

**The correct option is** $\mathrm{D}$**.**

**Q.15 Answer:**

$\mathrm{t}_1 = 2\mathrm{t}_2$

**Solution:**

**According to Kepler's law, the areal velocity of a planet around the sun always remains constant.**

$\mathrm{SCD} : \mathrm{A}_1 - \mathrm{t}_1$ **(areal velocity constant)** $\mathrm{SAB} : \mathrm{A}_2 - \mathrm{t}_2$

$\frac{\mathrm{A}_1}{\mathrm{t}_1} = \frac{\mathrm{A}_2}{\mathrm{t}_2}, \mathrm{t}_1 = \mathrm{t}_2 \cdot \frac{\mathrm{A}_1}{\mathrm{A}_2}, (\text{ given } \mathrm{A}_1 = 2\ \mathrm{A}_2)$

$= t_2 \cdot \frac{2A_2}{A_2} \quad \therefore \quad t_1 = 2t_2$

**Q.16 Answer:**

129

**Solution:**

**From Kepler's law of periods,**

$$\mathrm{T}_2 = \mathrm{T}_1 \left(\frac{\mathrm{R}_2}{\mathrm{R}_1}\right)^{3/2} = 365 \left(\frac{\mathrm{X}/2}{\mathrm{X}}\right)^{3/2}$$
$$= 365 \times \frac{1}{2\sqrt{2}} = 129 \text{ days.}$$

**Q.17 Answer:**

one-eighth the present year

**Solution:**

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$\frac{T_1^2}{T_2^2} = \frac{r_1^3}{r_2^3} = (4)^3 \quad \therefore T_2 = T_1/8.$

**Q.18 Answer:**

$10\sqrt{10}$

**Solution:**

$T^2 \propto R^3$ **(According to Kepler's law)** $T_1^2 \propto \left(10^{13}\right)^3$ **and** $T_2^2 \propto \left(10^{12}\right)^3$

$\therefore \quad \frac{T_1^2}{T_2^2} = (10)^3$ **or** $\frac{T_1}{T_2} = 10\sqrt{10}$

**Q.19 Answer:**

$1 : 2\sqrt{2}$

**Solution:**

**Time period does not depend on the mass. Also,** $T^2 \propto r^3$**.**

**Q.20 Answer:**

$\frac{12}{1}$

**Solution:**

$\because$ **total mechanical energy** $E = -\frac{GMm}{2r}$

$\therefore \frac{E_1}{E_2} = \frac{\mathrm{m}_1}{\mathrm{m}_2} \times \frac{r_2}{r_1} = \frac{3}{1} \times \frac{4r}{r} = \frac{12}{1}$

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